

Zicong Fan

DIGITAL HUMAN · HUMAN/HAND-OBJECT INTERACTION

Zürich Switzerland

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About Me

Humans engage with the world through various interactions with objects and environments. I am a doctoral student at ETH Zürich supervised by Professor **Otmar Hilliges** and Max-Planck Institute Director **Michael J. Black**. My research focuses on advancing digital human understanding, with a particular emphasis on full-body human-object interaction, hand-object interaction and hand pose estimation. I am also interested in other digital human areas.

Education

ETH Zürich

PH.D. IN COMPUTER SCIENCE

Supervised by ETH Zürich Professor **Otmar Hilliges** and Max-Planck Institute Director **Michael J. Black**

Switzerland

August 2020 - NOW

The University of British Columbia

M.SC. IN COMPUTER SCIENCE

Supervised by Professor **Leonid Sigal** and Professor **Jim Little**

Canada

September 2018 - June 2020

The University of British Columbia

B.SC. IN COMPUTER SCIENCE

Supervised by Professor **Ian M. Mitchell**

Canada

September 2016 - June 2018

Employment

Reality Labs Research @ Meta

RESEARCH SCIENTIST INTERN

Inverse rendering of hand avatar

Switzerland

June 2024 - NOW

Perceiving Systems @ Max Planck Institute for Intelligent Systems

GUEST SCIENTIST

Human-object interaction

Germany

April 2022 - NOW

Advanced Interactive Technologies @ ETH Zürich

GRADUATE RESEARCH ASSISTANT

Human-object interaction

Switzerland

August 2020 - NOW

Skills

Expertise 3D reconstruction, digital humans, implicit representations, motion capture

Proficient PyTorch, Python, LaTeX, Linux, Git, OpenCV, NumPy, SciPy

Moderate Premiere Pro, Blender, Photoshop

Prior experience Java, C/C++, MATLAB, ROS, HTML/CSS

Languages English, Cantonese, Mandarin

Publications

GraspXL: Generating Grasping Motions for Diverse Objects at Scale [🔗](#)

HUI ZHANG, SAMMY CHRISTEN, [ZICONG FAN](#), OTMAR HILLIGES, JIE SONG

ECCV 2024, Milano IT

Benchmarks and Challenges in Pose Estimation for Egocentric Hand Interactions with Objects [🔗](#)

[ZICONG FAN](#)^{*}, TAKEHIKO OHKAWA^{*}, LINLIN YANG^{*}, NIE LIN, ZHISHAN ZHOU, SHIHAI ZHOU, JIAJUN LIANG, ZHONG GAO, XUANYANG ZHANG, XUE ZHANG, FEI LI, LIU ZHENG, FENG LU, KARIM ABOU ZEID, BASTIAN LEIBE, JEONGWAN ON, SEUNGRYUL BAEK, ADITYA PRAKASH, SAURABH GUPTA, KUN HE, YOICHI SATO, OTMAR HILLIGES, HYUNG JIN CHANG, ANGELA YAO

ECCV 2024, Milano IT

HOLD: Category-agnostic 3D Reconstruction of Interacting Hands and Objects from Video



ZICONG FAN, MARIA PARELLI, MARIA ELENI KADOGLOU, MUHAMMED KOCABAS, XU CHEN, MICHAEL J. BLACK, OTMAR HILLIGES

CVPR 2024, Seattle US

Highlight (Top 11.9%) 🏆

ArtiGrasp: Physically Plausible Synthesis of Bi-Manual Dexterous Grasping and Articulation

HUI ZHANG, SAMMY CHRISTEN, ZICONG FAN, LUOCHENG ZHENG, JEMIN HWANGBO, JIE SONG, OTMAR HILLIGES

3DV 2024, Davos CH

Spotlight 🏆

ARCTIC: A Dataset for Dexterous Bimanual Hand-Object Manipulation



ZICONG FAN, OMID TAHERI, DIMITRIOS TZIONAS, MUHAMMED KOCABAS, MANUEL KAUFMANN, MICHAEL J. BLACK, OTMAR HILLIGES

CVPR 2024, Vancouver CA

HMP: Hand Motion Priors for Pose and Shape Estimation from Video



ENES DURAN, MUHAMMED KOCABAS, VASILEIOS CHOUTAS, ZICONG FAN, MICHAEL J. BLACK

WACV 2024, Waikoloa US

TempCLR: Reconstructing Hands via Time-Coherent Contrastive Learning



ANDREA ZIANI*, ZICONG FAN*, MUHAMMED KOCABAS, SAMMY CHRISTEN, OTMAR HILLIGES

3DV 2022, Prague CZ

Learning to Disambiguate Strongly Interacting Hands via Probabilistic Per-pixel Part Segmentation



ZICONG FAN, ADRIAN SPURR, MUHAMMED KOCABAS, SIYU TANG, MICHAEL J. BLACK, OTMAR HILLIGES

3DV 2021, Virtual

Oral Presentation 🏆

A Skeleton-driven Neural Occupancy Representation for Articulated Hands



KORRAWEE KARUNRATANAKUL, ADRIAN SPURR, ZICONG FAN, OTMAR HILLIGES, SIYU TANG

3DV 2021, Virtual

Oral Presentation 🏆

Learning Motion Predictors for Smart Wheelchair using Autoregressive Sparse Gaussian Process



ZICONG FAN, LILI MENG, TIAN QI CHEN, JINGCHUN LI, IAN M MITCHELL

ICRA 2018, Brisbane AU

Services

WORKSHOP ORGANIZERS

- HANDS2024** Observing and Understanding Hands in Action @ ECCV, Milano IT
- HANDS2023** Observing and Understanding Hands in Action @ ICCV, Paris FR

REVIEWER

- CVPR** 2023, 2024, 2025
- ICCV** 2023
- ECCV** 2022, 2024
- SIGGRAPH ASIA** 2023
- 3DV** 2023, 2024
- RA-L** 2023

Selected GitHub Repositories

arctic 320 stars | **hold** 335 stars

Honors & Awards

- 2019 **Alexander Graham Bell: Canada Graduate Scholarship-Master's**, NSERC Canada
- 2017 **Undergraduate Student Research Awards**, NSERC Canada
- 2016 **the Governor General's Collegiate Bronze Medal**, the Governor General of Canada